Shubham Choudhury

PhD in Computational Biology

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Education

Indraprastha Institute of Information Technology, Delhi (IIIT-D) **Jan 2021 – July 2026**

Doctor of Philosophy in Computational Biology | CGPA: 8.29 / 10.00 *New Delhi, India*

- **Thesis:** Development of AI-Based Tools for Functional Annotation of lncRNA and Its Applications
- **Supervisor:** Prof. Gajendra P.S. Raghava (Head, Computational Biology)
- Awarded the prestigious **DST-INSPIRE Fellowship** for doctoral research.

Indian Institute of Science Education and Research, Kolkata (IISER) **July 2014 – July 2019**

BS-MS Dual Degree in Biological Sciences | CGPA: 7.11 / 10.00 *Kolkata, India*

- **Master's Thesis:** Transcriptomic Meta-Analysis of Human Sepsis and Related Illnesses
- Qualified **DBT-JRF (Category I)**, awarded by the Dept. of Biotechnology, Govt. of India.

Research Experience

CSIR – Institute of Microbial Technology, Chandigarh (IMTECH) **July 2019 – July 2020**

Project Assistant | *Supervisor: Dr. Manoj Kumar* *Chandigarh, India*

- Built **CoronaVR**, a web resource consolidating epitope and therapeutic data for SARS-CoV-2 (*Frontiers in Microbiology*, 2020).
- Conducted metagenomic sequencing and virome analysis of sewage samples across Chandigarh to track environmental spread of viral pathogens.
- Performed computational drug repurposing for virus-associated diseases via drug–target network analysis, contributing to a publication in *Computers in Biology and Medicine* (2021).

Key Research Projects

lncrnaPI — LLM-Based Prediction of lncRNA–Protein Interacting Pairs [🌐] **Nov 2025**

Python · DNABERT-2 · ESM-2 · CatBoost · CNN · BLAST · Nfeature · Pfeature · POSSUM · Web Server Development

- Curated the largest non-redundant LPI dataset to date (524,488 pairs from NPInter v5.0); generated features using DNABERT-2, ESM-2-t30, PSSM profiles, and compositional descriptors.
- Best model (CatBoost + LLM embeddings) achieved **AUC = 0.989, MCC = 0.915**, substantially outperforming the only existing functional tool (LPI-BLS, AUC = 0.512).
- Deployed a web server and released `lncrnaPI` as a Python package on PyPI for genome-scale lncRNA–protein interaction scanning.

LLM-Based Prediction of Pancreatic Cancer from Blood Exosomal Transcriptomics **June 2024**

Python · PeptideBERT · ProtBERT · ESM2 · BLAST · MERCI · Scikit-learn · PyTorch

- Mapped exosomal gene expression profiles from 501 PDAC/non-PDAC patients into protein sequence representations.
- Fine-tuned three protein language models (PeptideBERT, ProtBERT, ESM2) for binary cancer classification.
- Best model (ESM2-t6_8M_UR50D) achieved **AUROC = 0.95** on an independent held-out dataset.
- Developed an ensemble combining LLM-based predictions with alignment-based approaches (BLAST, MERCI).

CytoLNCpred — Subcellular Localization of lncRNA Across 15 Cell Lines [🌐] **Jan 2023**

Python · Machine Learning · DNABERT-2 · Web Server Development

- Curated lncRNA localization data from lncAtlas; extracted *k*-mer composition, correlation, and DNABERT-2 embedding features.
- Trained and benchmarked classical ML models and a transformer-based LLM; achieved **mean AUROC = 0.709** across 15 cell lines.
- Deployed a publicly accessible prediction web server; published in *Frontiers in Bioinformatics* (2025).

MRSLpred — Multi-Label Subcellular Localization of Human mRNA [🌐] **Jan 2022**

Python · Machine Learning · NFeature · Web Server Development

- Built a genome-scale predictor for six mRNA localization compartments using features from the RNALocate database.
- Boosted accuracy via MERCI motif features; achieved **mean AUROC = 0.742**. Published in *Frontiers in Bioinformatics* (2024).

Transcriptomic Meta-Analysis of Human Sepsis (Master's Thesis) **May 2018 – June 2019**

R · RStudio · Bash · HPC Computing

- Analyzed public sepsis microarray datasets from GEO to identify differentially expressed genes and pathways.

- Assessed whether upregulation of the Osteoclast Differentiation Pathway is sepsis-specific or a generic inflammatory response.

Teaching Experience

Indraprastha Institute of Information Technology, Delhi (IIIT-D) **Winter 2022 – Winter 2024**
Teaching Assistant, Department of Computational Biology *New Delhi, India*

- BIO543: Big Data Mining in Healthcare (Winter 2022, Winter 2024)
- BIO542: Machine Learning for Biomedical Applications (Monsoon 2022)
- Assisted with tutorial sessions, assignment design, grading, and student mentoring for graduate-level courses in machine learning and biomedical data science.

Technical Skills

Programming & Scripting: Python, R, Bash, PHP
Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, Keras; Transformers/LLMs (DNABERT-2, ESM2, ProtBERT)
Bioinformatics Libraries: Pandas, NumPy, Bioconductor, Biopython
Databases & Web: MySQL, REST APIs, Apache, HTML5; experience deploying bioinformatics web servers
DevOps & HPC: Git, GitHub, Docker, HPC/SLURM cluster computing
Specialised Domains: lncRNA biology, RNA subcellular localization, RNA–protein interaction, environmental viromics, LLMs for genomics
Wet-Lab Techniques: RNA/DNA extraction, PCR, gel electrophoresis, viral enrichment, RNA-seq library preparation (Nanopore)

Publications

First-Author Publications

[1] **S. Choudhury**, N. Bajiya, G.P.S. Raghava. "Prediction of lncRNA-protein interacting pairs using LLM embeddings based on evolutionary information." *bioRxiv* (2025). **[Preprint; Under Review]** [DOI]

[2] **S. Choudhury**, N.K. Mehta, G.P.S. Raghava. "CytoLNCpred: A computational method for predicting cytoplasm-associated long non-coding RNAs in 15 cell lines." *Frontiers in Bioinformatics* **5** (2025): 1585794. [DOI]

[3] **S. Choudhury**, A.S. Rathore, G.P.S. Raghava. "Compilation of resources on subcellular localization of lncRNA." *Frontiers in RNA Research* **2** (2024): 1419979. [DOI]

[4] **S. Choudhury**, N. Bajiya, S. Patiyal, G.P.S. Raghava. "MRSLpred — a hybrid approach for predicting multi-label subcellular localization of mRNA at the genome scale." *Frontiers in Bioinformatics* **4** (2024): 1341479. [DOI]

Co-Authored Publications

[5] S. Aggarwal, A. Dhall, S. Patiyal, **S. Choudhury**, A. Arora, G.P.S. Raghava. "An ensemble method for prediction of phage-based therapy against bacterial infections." *Frontiers in Microbiology* **14** (2023): 1148579. [DOI]

[6] N. Bajiya, **S. Choudhury**, A. Dhall, G.P.S. Raghava. "AntiBP3: A method for predicting antibacterial peptides against Gram-positive/negative/variable bacteria." *Antibiotics* **13**(2) (2024): 168. [DOI]

[7] A. Dhall, S. Patiyal, **S. Choudhury**, S. Jain, K. Narang, G.P.S. Raghava. "TNFepitope: A webserver for the prediction of TNF-α inducing epitopes." *Computers in Biology and Medicine* **160** (2023): 106929. [DOI]

[8] P.S. Gahlot, **S. Choudhury**, N. Bajiya, N. Kumar, G.P.S. Raghava. "Prediction of plant resistance proteins using alignment-based and alignment-free approaches." *Proteomics* **25**(5–6) (2025): e202400261. [DOI]

[9] A.K. Gupta, M.S. Khan, **S. Choudhury**, et al. "CoronaVR: A computational resource and analysis of epitopes and therapeutics for SARS-CoV-2." *Frontiers in Microbiology* **11** (2020): 1858. [DOI]

[10] N. Kumar, **S. Choudhury**, N. Bajiya, S. Patiyal, G.P.S. Raghava. "Prediction of anti-freezing proteins from their evolutionary profile." *Proteomics* **25**(3) (2025): e202400157. [DOI]

[11] N. Kumar, S. Patiyal, **S. Choudhury**, R. Tomer, A. Dhall, G.P.S. Raghava. "DMPPred: A tool for identification of antigenic regions responsible for inducing Type 1 Diabetes Mellitus." *Briefings in Bioinformatics* **24**(1) (2023): bbac525. [DOI]

[12] A. Mukhopadhyay, **S. Choudhury**, M. Kumar. "Metaviromic analyses of DNA virus community from sediments of the N-Choe Stream, North India." *Virus Research* **330** (2023): 199110. [DOI]

[13] A. Rajput, A. Thakur, A. Rastogi, **S. Choudhury**, M. Kumar. "Computational identification of repurposed drugs against viruses causing epidemics and pandemics via drug–target network analysis." *Computers in Biology and Medicine* **136** (2021): 104677. [DOI]

[14] A.S. Rathore, **S. Choudhury**, A. Arora, P. Tijare, G.P.S. Raghava. "ToxinPred 3.0: An improved method for predicting the toxicity of peptides." *Computers in Biology and Medicine* **179** (2024): 108926. [DOI]

[15] A.S. Rathore, N. Kumar, **S. Choudhury**, N.K. Mehta, G.P.S. Raghava. "Prediction of hemolytic peptides and their hemolytic concentration." *Communications Biology* **8**(1) (2025): 176. [DOI]

Competitions and Challenges

CODA TB DREAM Challenge — 3rd Place (International Team Competition) **March 2024**
Machine Learning · Audio Signal Processing · Tuberculosis Screening · Digital Health

- Collaborated with lab members in the international **CODA TB DREAM Challenge**, focused on AI-based tuberculosis screening using cough audio recordings and clinical metadata.
- Developed predictive machine learning models for identifying TB-positive individuals from elicited cough sounds collected across 7 countries.
- Secured **3rd position globally** among participating teams in the challenge organized under the DREAM Challenges framework.
- Challenge findings and benchmark results were published in:
  "CODA TB DREAM Challenge: Predicting Tuberculosis from Cough Sounds"

Honors and Awards

DST-INSPIRE Fellowship | Dept. of Science & Technology, Govt. of India **Jan 2021**
- Prestigious national fellowship awarded to top PhD students across India for outstanding academic performance.

DBT-JRF (Category I) | Dept. of Biotechnology, Govt. of India **June 2020**
- Awarded Category I rank under the DBT-Junior Research Fellowship program via the BET national entrance examination.

Conferences and Workshops

EMBO Workshop: Non-coding RNA Medicine **May 2023**
- Attended an international EMBO workshop on RNA therapeutics and non-coding RNA biology held in Poznań, Poland.

Data Dynamics Summit (DDS 2024), IISER Pune **March 2024**
- Participated in a national-level summit focusing on data science, computational biology, and interdisciplinary analytics.

IBAB Bio-IT Workshop: Introductory Omics and RNA-seq Data Analysis **July 2024**
- Completed a 3-day hands-on workshop on RNA-seq analysis, omics data processing, and bioinformatics pipelines at IBAB, Bangalore.

Symposium on Computational Gastronomy, IIIT Delhi **Dec 2023**
- Participated in an interdisciplinary symposium exploring computational approaches to food science and data-driven gastronomy.

Symposium on Computational Gastronomy, IIIT Delhi **Dec 2022**
- Attended symposium sessions on computational modeling and data analysis in food systems.

Additional Information

Languages: English (C1 – Advanced) · Hindi (B2 – Upper-Intermediate) · Odia (C2 – Native Proficiency)

References

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